

Deriv:
 $\frac{d}{dx}(x^2y+x) = (x^2y^2+x^2+y^2+1)dy \rightarrow$ qizim

$$\frac{x(y+1)dx}{(y^2+1)(x^2+1)} = \frac{(x^2+1) \cdot (y^2+1)dy}{(x^2+1) \cdot (y^2+1)}$$

$$\frac{x}{x^2+1} dx = \frac{y^2+1}{y+1} dy \quad \text{integratimi alabiriz.}$$

$$\frac{1}{2} \int \frac{2x}{x^2+1} dx + \int \left(y-1 + \frac{2}{y+1} \right) dy \rightarrow \frac{1}{2} \ln(x^2+1) = \frac{y^2}{2} y + 2 \ln(y+1) + C$$

Drnek:

$$y' - xy^2 + x = 0$$

$$y' = xy^2 - x$$

$$y' = x \cdot (y^2 - 1)$$

$$\frac{dy}{dx} = x \cdot (y^2 - 1)$$

$$\frac{dy}{(y^2-1)} = dx \cdot x \cdot \frac{(y^2-1)}{y^2-1} \quad \left. \vphantom{\frac{dy}{(y^2-1)}} \right\} \text{gerisini al}$$

$$\frac{y-1}{y+1} = e^{x^2}$$

$$(x+y)dx + dy = 0 \rightarrow \text{Ddeu}$$

$$x+y=u$$

$$dx+dy=du$$

$$dxu + du - dx = 0$$

$$du = dxu + dx$$

$$\frac{du}{(1-u)} = dx$$

$$\ln(1-u) = x+C$$

$$1-u = e^{x+C}$$

$$u = x+y$$

$$1-x-y = e^{x+C}$$

$$y = 1-x - e^{x+C}$$